Jonas Sénizergues

Computer Scientist

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https://jsenizer.github.io/

Research interests

Graph theory
Graph algorithms
Distributed algorithms
Autostabilization
Byzantine fault tolerance
Game theory

Employment

2023-2024 **Teaching Assistant**, *Université de Bordeaux*
2021-2022 **Teaching Assistant**, *ENSIIE*
2018-2021 **PhD student**, *Université Paris-Saclay*
2016-2017 **Math Teacher in highschool**, *Bondoufle*

Diplomas

2022 **PhD**, *Computer Science*, Université Paris-Saclay
2015 **ENS diploma**, ENS Cachan
2014 **Agrégation de Mathématiques**, *(a prestigious mathematics competitive exam in France)*
2013 **Master**, *Computer Science*, MPRI
2011 **Licence**, *Computer Science*, ENS Cachan

Curriculum

2018–2022 **PhD**, *LISN*, Université Paris-Saclay, Orsay
2014–2015 **Master BIBS (as auditor)**, *Université Paris-Sud*, Orsay
2013–2014 **Training for the Agrégation de Mathématiques competitive exam**, *ENS Cachan*, Cachan
2011–2013 **Master in Computer Science**, *MPRI*, Paris
2010–2013 **Computer Science**, *ENS Cachan*, Cachan
2008–2010 **CPGE**, *Lycée Michel Montaigne*, Bordeaux

Languages

French Native speaker
English Fluent
Spanish B1
Japanese A2

Programming languages

- ssr-Coq
- OCaml
- Python
- Java

Publications

International Conference

2026 **Maker-Maker games of rank 4 are PSPACE-complete**, *Florian Galliot, Jonas Sénizergues*, STACS 2026 , https://hal.science/hal-05060987/

2025 **A unified convention for achievement positional games**, *Florian Galliot, Jonas Sénizergues*, EUROCOMB2025 , https://hal.science/hal-05002388/

2023 **Making Self-Stabilizing Algorithms for any Locally Greedy Problem**, *Johanne Cohen, Laurence Pilard, Mikaël Rabie and Jonas Sénizergues*, SAND 2023

2021 **Self-Stabilization and Byzantine Tolerance for Maximal Independent Set**, *Johanne Cohen, Laurence Pilard and Jonas Sénizergues*, SSS 2021

Journal

2014 **Formalization of Shannon's Theorems**, *Reynald Affeldt, Manabu Hagiwara, and Jonas Sénizergues*, JAR 2014

Software Library

2019 **A Coq formalization of information theory and linear error correcting codes**, *Reynald Affeldt, Manabu Hagiwara, Jonas Sénizergues, Jacques Garrigue, Kazuhiko Sakaguchi, Taku Asai, Takafumi Saikawa, Naruomi Obata and Alessandro Bruni*

Preprint

preprint **A two-player version of the assignment problem**, *Florian Galliot, Nacim Oijid, Jonas Sénizergues*, submitted to IWOCA 2026 , https://arxiv.org/abs/2602.02628

preprint **Self-Stabilization and Byzantine Tolerance for Maximal Independent Set (journal ver.)**, *Johanne Cohen, Laurence Pilard, François Pirot and Jonas Sénizergues*, To be submitted, https://arxiv.org/abs/2210.06116v3

preprint **Autostabilizing Minimal Clique Decomposition with Byzantine Fault Tolerance**, *Johanne Cohen, Laurence Pilard and Jonas Sénizergues*, To be submitted, https://hal.science/hal-04872001

preprint **Minimum Colored Maximum Matchings in vertex-colored Graphs**, *Johanne Cohen, Yannis Manoussakis and Jonas Sénizergues*, To be submitted, https://hal.science/hal-02157745/

PhD Manuscript

2022 **Independent sets and beyond, through the prism of distributed systems and colored graphs**, *Jonas Sénizergues*, https://theses.hal.science/tel-03956035